

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Concept Mapping

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
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Section / Batch:	2/2025	Date:	08/07/2026

Code of Conduct

I C.MANOJ (192572085) _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: c.manoj _____

Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

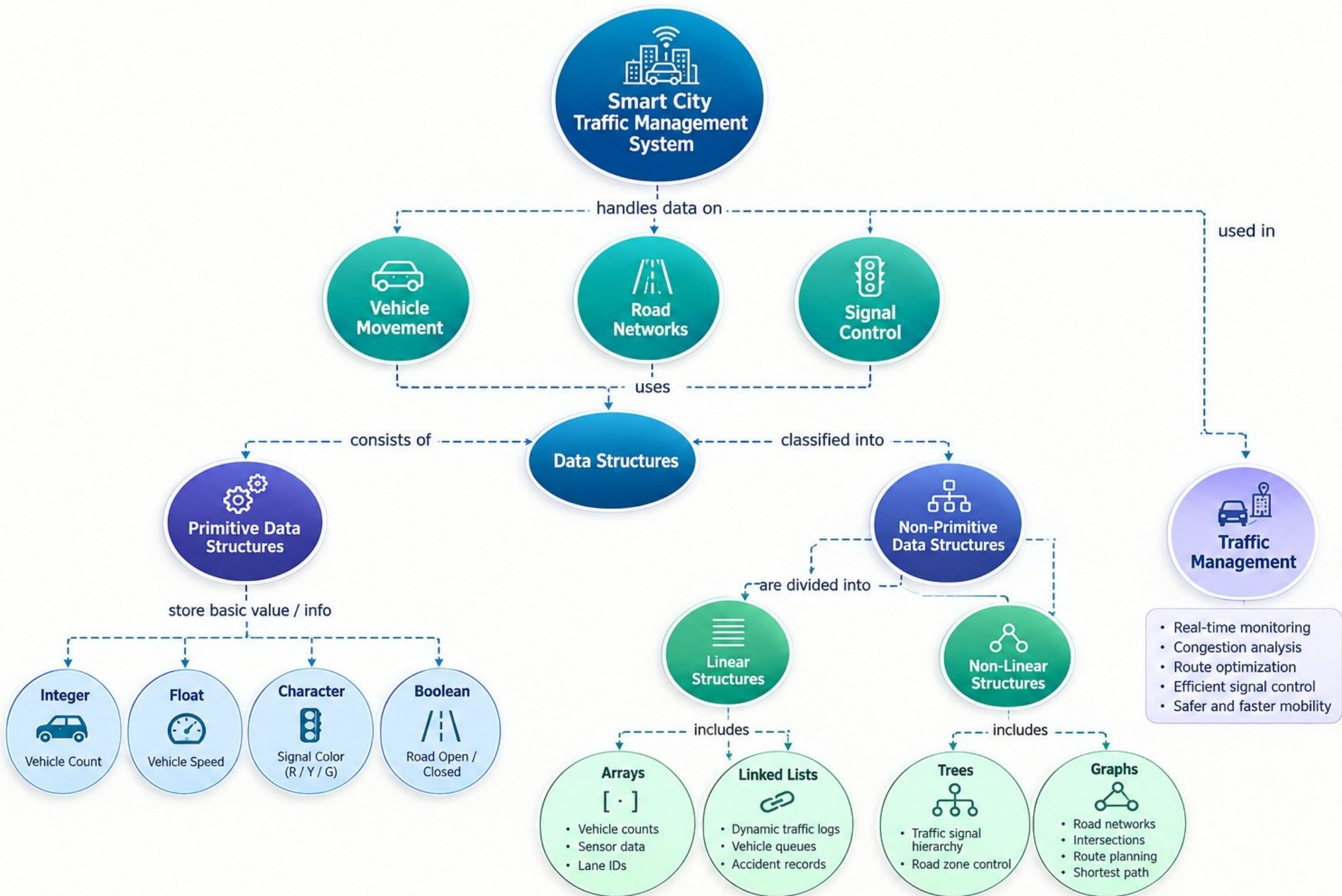
Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	2	20
Search Data Structure	Linear Search and Binary Search	3	20
Recursion, Performance analysis	Recursion	4	20
Tree Data Structures	Classifications of Tree data structure	5	20
GRAND TOTAL:			100 Marks

Annexure A – Concept Mapping

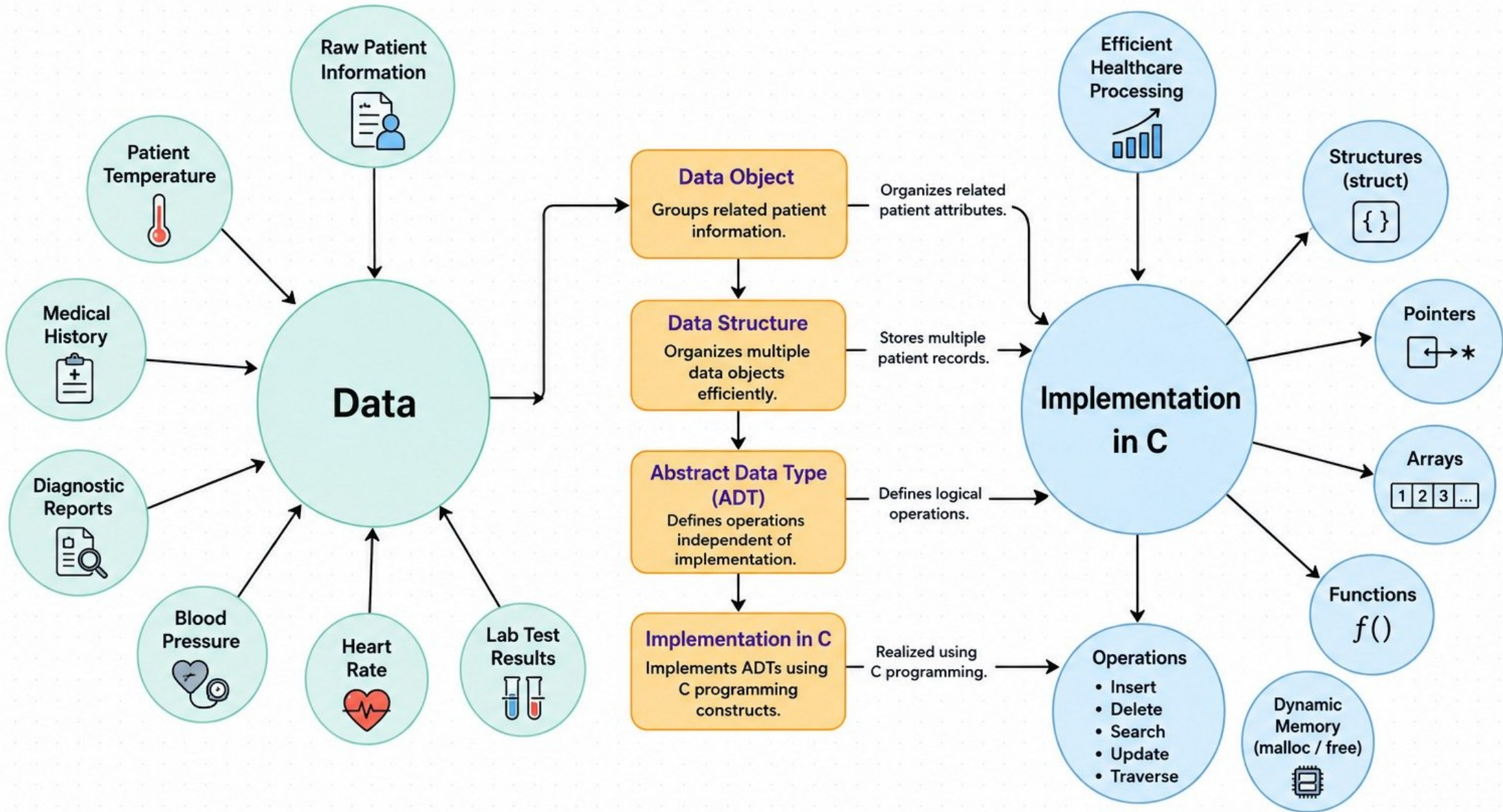
1. A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency. (20 Marks)
2. A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance. (20 Marks)
3. E-commerce platform manages product storage, searching, and recommendation systems. Construct a concept map linking Linear Data Structures → Sequential Access → Limitations → Non-Linear Data Structures → Hierarchical/Network Relationships → Applications. Analyze the differences between linear and non-linear structures and demonstrate how each supports specific components of the system. Justify which structure is more efficient for handling complex applications. (20 Marks)
4. A video streaming application must balance speed and memory usage. Construct a concept map connecting Time Complexity → Space Complexity → Tradeoff → Algorithm Design → Performance Optimization. Analyze how improving one factor affects the other and justify the optimal balance. (20 Marks)
5. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand



Concept Map: Smart Healthcare System – Data Organization



Outcome / Benefits

This hierarchical organization improves data storage, enables faster processing of patient records, supports efficient searching and updating, reduces memory usage, and enhances the overall performance of the smart healthcare system.

E-COMMERCE PLATFORM

Product Storage, Searching & Recommendation Systems

1

LINEAR DATA STRUCTURES

Elements arranged in a sequential manner

Array

Linked List

Stack

Queue

2

SEQUENTIAL ACCESS

Access elements one after another

- Easy traversal in one direction
- Suitable for simple operations
- Used in product lists, carts, order lists

3

LIMITATIONS

Not efficient for complex relationships

- Linear search is time-consuming ($O(n)$)
- Insertion/Deletion may be costly
- Poor performance for hierarchical or many-to-many relationships

LINEAR vs NON-LINEAR

LINEAR	NON-LINEAR
Sequential	Non-Sequential
Simple relationships	Complex relationships
Easy to implement	More complex
Less efficient for large & complex data	More efficient for large & complex data
Examples: Array, Linked List, Stack, Queue	Examples: Tree, Graph, Hash Table, Trie



CONCLUSION

Linear structures are easy and useful for simple, sequential operations. Non-linear structures are more efficient and scalable for handling complex relationships in real-world e-commerce applications.

4

NON-LINEAR DATA STRUCTURES

Elements not arranged sequentially

Tree

Graph

Hash Table

Trie

5

HIERARCHICAL / NETWORK RELATIONSHIPS

Supports complex connections & relationships

- Tree → Hierarchical (Parent-Child)
- Graph → Many-to-Many relationships
- Hash Table → Fast search (Key-Value)
- Trie → Efficient prefix-based search

6

APPLICATIONS IN E-COMMERCE

Real-world use cases



Product Search → Trie, Hash Table (fast lookup)



Product Categories → Tree (hierarchical structure)



Recommendation System → Graph (user-product relations)



Inventory & Warehouse → Graph, Queue (workflow)

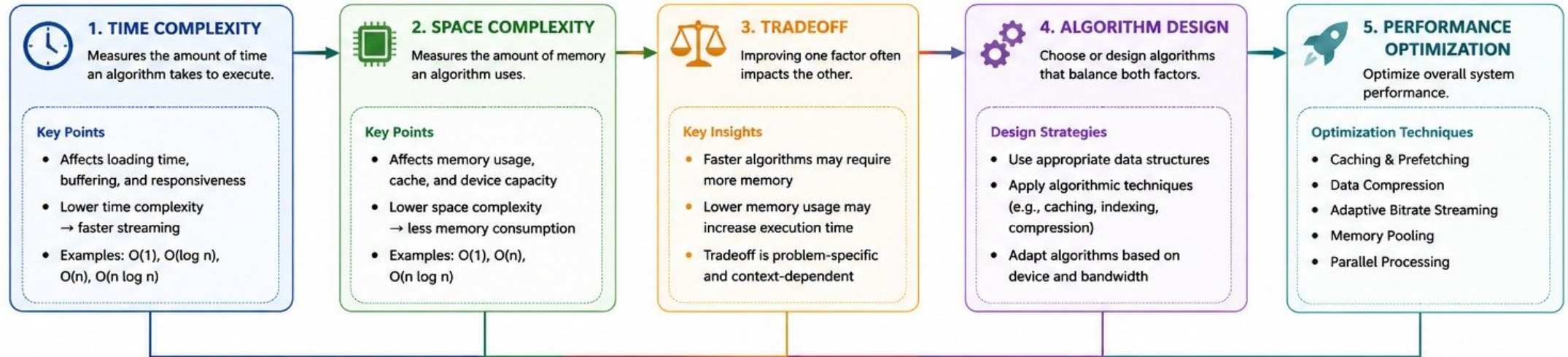


JUSTIFICATION

For an e-commerce platform with complex product data, user behavior, and recommendation needs, non-linear data structures are more efficient, scalable, and better suited for real-time performance.

VIDEO STREAMING APPLICATION

Balancing Speed and Memory Usage



OPTIMAL BALANCE

Achieve the best tradeoff between time and space for smooth, efficient, and scalable streaming.

HOW IMPROVING ONE AFFECTS THE OTHER



Improving Time (Faster Execution)

Often requires more memory
Example: Preloading, caching, indexing



↑ Space Complexity
(More Memory Used)



Improving Space (Less Memory)

Often increases execution time
Example: Computing on-demand, no caching



↑ Time Complexity
(Slower Execution)

Key Idea: You can't minimize both simultaneously; the goal is to find the right balance.

EXAMPLES IN VIDEO STREAMING

Technique	Time Complexity	Space Complexity	Use Case
Adaptive Bitrate Streaming	Lower (Fast Switching)	Higher (Multiple Quality Files)	Smooth playback on varying networks
Video Caching	Lower (Quick Access)	Higher (Stored Data in Cache)	Reduce buffering and latency
On-demand Processing	Higher (Compute at Playback)	Lower (Less Memory Used)	Low-end devices with limited memory
Compression Algorithms	Higher (Compression Time)	Lower (Smaller Data Size)	Save bandwidth and storage

JUSTIFICATION OF OPTIMAL BALANCE



User Experience: Fast startup, no buffering, high-quality playback.



Device Constraints: Limited memory on mobile and IoT devices.



Network Conditions: Varying bandwidth requires adaptive strategies.



Scalability: Efficient resource usage ensures cost-effective scaling.

Thus, the optimal balance ensures maximum performance with minimum resource usage.



CONCLUSION

In a video streaming application, achieving the right balance between Time Complexity and Space Complexity is crucial. By carefully designing algorithms and applying optimization techniques, we can deliver high performance with efficient memory usage, ensuring a smooth and scalable streaming experience.

DIGITAL ARCHIVE

Stores Millions of Documents

SEARCHING

Process of finding a document in the archive

can be done using

LINEAR SEARCH

Checks each item one by one

works on

UNSORTED / ANY ORDERED DATA
No requirement of sorting

time complexity

$O(n)$

- Worst Case: $O(n)$
- Average Case: $O(n)$
- Best Case: $O(1)$

performance

SLOW

Not suitable for large datasets
(Millions of documents)

improved using

BINARY SEARCH

Divides data into halves and searches

requires

SORTED DATA

Data must be in ascending/descending order

time complexity

$O(\log n)$

- Worst Case: $O(\log n)$
- Average Case: $O(\log n)$
- Best Case: $O(1)$

performance

FAST

Highly efficient for large datasets

IMPACT OF DATA ORGANIZATION

analysis

- Unsorted data → Only Linear Search possible → More comparisons → Slower.
- Sorted data → Binary Search possible → Fewer comparisons → Faster.
- Proper data organization (sorting + indexing) greatly improves search efficiency.

TIME COMPLEXITY

Measures efficiency of search

determines

PERFORMANCE

Overall speed and resource usage

Better Time Complexity
→ Higher Performance
(Less time taken)

Poor Time Complexity
→ Lower Performance
(More time taken)

COMPARISON

Feature	Linear Search	Binary Search
Data Requirement	Any order	Sorted data
Method	Sequential checking	Divide and conquer
Time Complexity (Average)	$O(n)$	$O(\log n)$
Suitable For	Small datasets	Large datasets
Performance	Low	High

JUSTIFICATION – BEST TECHNIQUE



For a digital archive with millions of documents, Binary Search is the best technique because:

- It has logarithmic time complexity $O(\log n)$, which scales very well.
- Requires fewer comparisons, reducing search time drastically.
- When combined with sorting and indexing, it ensures optimal performance.
- Therefore, organizing data in sorted order and using Binary Search provides the most efficient solution.